Gaussian mixture model is a type of clustering model that assumes the instances are generated from a mixture of Gaussian Distributions.

Compare with K-means

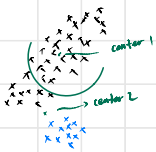
Instances:



Gaussian Mixture model



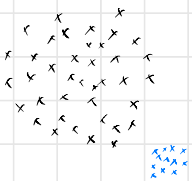
K-means



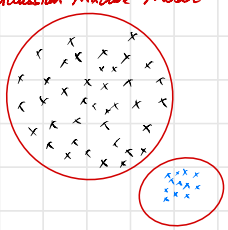
K-means Cannot handle elliptical clusters well

because it creats circles or balls around its centers

Instances:



Gaussian Mixture Model





K-means Cannot deal with clusters with different Variance well

Because it uses the same distance measure (normally L_2) for different clusters

Assumptions of Gaussian Mixture model

↗ Likelihood of observations

$$pr(x_1, \dots, x_n) = pr(x_1) \cdots pr(x_n)$$ independent assumption

$$= \prod_{i=1}^{n} \sum_{j=1}^{K} p(x_i \in \text{Cluster } j)\, p(x_i \mid x_i \in \text{Cluster } j)$$ Law of total probability

$$= \prod_{i=1}^{n} \sum_{j=1}^{K} \lambda_j N(x_i; \mu_j, \sigma_j^2)$$ Gaussian Assumption

Generation Process:

Latent Variable → (equals) goes to each cluster with probability → within each cluster, generate from a corresponding Gaussian Distribution → observation



The latent variable z is a discrete variable equal i (where $i = 1, \dots, k$) with probability λ_i. Given this latent variable, the likelihood of the data x is normally distributed with mean $= \mu_i$, Var $= \sigma_i^2$.

It is hard to maximize the likelihood function

$$pr(x_1, \dots x_n) = \prod_{i=1}^{n} \sum_{j=1}^{k} \lambda_j \mathcal{N}(x_i ; \mu_j, \sigma_j^2)$$

directly.

↓ minimize negative log-likelihood

$$\min_{\lambda, \mu, \sigma} (-l(\lambda, \mu, \sigma ; x_1, \dots, x_n)) = \min_{\lambda, \mu, \sigma} -\sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} \lambda_j \mathcal{N}(x_i ; \mu_j, \sigma_j^2)\right)$$

$$= \min_{\lambda, \mu, \sigma} -\sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} \lambda_j \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(x_i - \mu_j)^T \Sigma_j^{-1} (x - \mu_j)\right\}\right)$$

→ we use σ^2 to denote the parameter of covariance

S.t. $\sum_{j=1}^{k} \lambda_j = 1$

Σ_j is symmetric and positive semi definite

because of this sum, we cannot use log to cancel out exp

there is no analytical form if we take gradient wrt λ, μ, σ and set the gradients to 0

need to satisfy the constraints,

So directly use gradient descent is not ideal

☆ The problem is also ill-defined. If $\mu_j \to x_i$, $\Sigma_j \to 0$, then $pr(x_1, \dots, x_n) \to \infty$

Below we consider how to use E-M algorithm to solve these issues.

E-M algorithm

problem: $\underset{\theta}{\text{argmax}}\ l(\theta ; x)$ → this is hard

Observation: If we introduce some latent variables z, then

$\underset{\theta}{\text{argmax}}\ l(\theta ; x, z)$ → this is easy

E-M algorithm:

$$\theta^{(t+1)} = \underset{\theta}{\text{argmax}} \underset{z \sim p(\cdot | x, \theta^{(t)})}{E} l(\theta ; x, z)$$

M-Step

→ E-Step

Why does E-M algorithm work?

Notice that

$$p(x, z; \theta) = p(x; \theta) p(z|x; \theta)$$

$$\Rightarrow \ell(\theta; x, z) = \ell(\theta; x) + \ell(\theta; z|x)$$

$$\Rightarrow \ell(\theta; x) = \ell(\theta; x, z) - \ell(\theta; z|x)$$

↳ target ↳ the one we can maximize easily

$$\Rightarrow \ell(\theta; x) = \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; z|x))$$

take integration on both sides

$$\Rightarrow \ell(\theta^{(t)}; x) = \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z)) - \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; z|x))$$

$$\Rightarrow \ell(\theta; x) - \ell(\theta^{(t)}; x) = \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z))$$

$$- \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; z|x)) + \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; z|x))$$

$$= \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z)) - \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z))$$

$$+ KL(p(z|x, \theta^{(t)}), p(z|x, \theta)) \quad \rightarrow \geq 0$$

Since

$$\theta^{(t+1)} = \arg\max_{\theta} \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta; x, z))$$

We have

$$\mathop{E}_{z \sim p(\cdot|x, \theta^{(t+1)})} (\ell(\theta^{(t+1)}; x, z)) \geq \mathop{E}_{z \sim p(\cdot|x, \theta^{(t)})} (\ell(\theta^{(t)}; x, z))$$

So $\ell(\theta^{(t+1)}; x) \geq \ell(\theta^{(t)}; x)$, i.e. E-M algorithm leads to non-decreasing likelihood function at each step.

E-M algorithm for Gaussian Mixture model

Introduce latent variables: $z_{ij} = \begin{cases} 1 & \text{if } x_i \in C_j \\ 0 & \text{otherwise} \end{cases}$

In order to use E-M algorithm, we need joint distribution $p(x, z; \theta^{(t)})$ and conditional distribution $p(z \mid x; \theta^{(t)})$

E - Step:

Joint distribution

$$p(x, z; \lambda, \mu, \Sigma) = \prod_{i=1}^{n} \prod_{j=1}^{k} \left(\lambda_j \mathcal{N}(x_i; \mu_j, \Sigma_j) \right)^{z_{ij}}$$

$$\Rightarrow l(\lambda, \mu, \Sigma; x, z) = \sum_{i=1}^{n} \sum_{j=1}^{k} \left(z_{ij} \left(\log \lambda_j + \log \mathcal{N}(x_i; \mu_j, \Sigma_j) \right) \right)$$

Conditional distribution

$$p(z_{ij}=1 \mid x; \lambda^{(t)}, \mu^{(t)}, \Sigma^{(t)}) = p(z_{ij}=1 \mid x_i, \lambda^{(t)}, \mu^{(t)}, \Sigma^{(t)})$$

$$= \frac{p(z_{ij}=1; \lambda^{(t)}) \, p(x_i \mid z_{ij}=1; \lambda^{(t)}, \mu^{(t)}, \Sigma^{(t)})}{\sum_{j=1}^{K} p(z_{ij}=1; \lambda^{(t)}) \, p(x_i \mid z_{ij}=1; \lambda^{(t)}, \mu^{(t)}, \Sigma^{(t)})} \quad \text{Bayes Theorem}$$

$$= \frac{\lambda_j^{(t)} \mathcal{N}(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}{\sum_{j=1}^{K} \lambda_j^{(t)} \mathcal{N}(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}$$

$$= p_{ij}^{(t)}$$

So we have

$$\underset{z \sim p(\cdot \mid x, \lambda^{(t)}, \mu^{(t)}, \Sigma^{(t)})}{E} \left(l(\lambda, \mu, \Sigma; x, z) \right) = \sum_{i=1}^{n} \underset{z \sim p(\cdot \mid x^{(t)}, \mu^{(t)}, \Sigma^{(t)})}{E} \left(l(\lambda, \mu, \Sigma; x_i, z_i) \right)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{k} \left(p_{ij}^{(t)} \left(\log \lambda_j + \log \mathcal{N}(x_i; \mu_j, \Sigma_j) \right) \right)$$

↓ notice that we can maximize them independently ↓

first part: maximize λ

$$\underset{\lambda_1,\dots,\lambda_K}{\operatorname{argmax}} \sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)} \ln \lambda_j$$

s.t. $0 \le \lambda_j \le 1$ → this is not necessary. Since we are dealing with $\ln$'s $\Rightarrow \lambda_j > 0$ and $\lambda_1 + \dots + \lambda_K = 1$ $\Rightarrow \lambda_j \le 1$, so the result will satisfy this automatically if exist

$\lambda_1 + \dots + \lambda_K = 1$

Lagrange form:

$$\mathcal{L}(\lambda_1,\dots,\lambda_K,\lambda) = \sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)} \ln \lambda_j + \lambda(\lambda_1+\dots+\lambda_K-1)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_j} = \frac{\sum_{i=1}^{n} p_{ij}^{(t)}}{\lambda_j} + \lambda = 0$$

$$\Rightarrow \lambda_j = \frac{\sum_{i=1}^{n} p_{ij}^{(t)}}{-\lambda} = \frac{\sum_{i=1}^{n} p_{ij}^{(t)}}{\sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)}}$$

put $\lambda_1+\dots+\lambda_K=1$ in, so we have $\frac{\sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)}}{-\lambda} = 1$

Second part: maximize μ and Σ

$$\underset{\mu_j,\Sigma_j}{\operatorname{argmax}} \sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)} \ln \mathcal{N}(x_i;\mu_j,\Sigma_j)$$

$\mathcal{N}(x;\mu,\Sigma) = (2\pi)^{-\frac{d}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

$$\Leftrightarrow \underset{\mu_j,\Sigma_j}{\operatorname{argmax}} \sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^{(t)} \left(-\frac{d}{2}\ln(2\pi) - \frac{1}{2}\ln(\det(\Sigma_j)) - \frac{1}{2}(x_i-\mu_j)^T\Sigma_j^{-1}(x_i-\mu_j)\right)$$

$\frac{\partial}{\partial A}\ln(\det(A)) = (A^{-1})^T$

$\frac{\partial}{\partial x}(x^TAx) = (A+A^T)x$

$\frac{\partial}{\partial A}(x^TAx) = xx^T$

$\frac{\partial}{\partial A}(x^TA^{-1}x) = -A$

take derivative wrt μ_j

$$\Rightarrow \sum_{i=1}^{n} p_{ij}^{(t)} 2\Sigma_j^{-1}(x_i-\mu_j) = 0$$

$$\Rightarrow \sum_{i=1}^{n} p_{ij}^{(t)}\Sigma_j^{-1}x_i = \sum_{i=1}^{n} p_{ij}^{(t)}\Sigma_j^{-1}\mu_j$$

$|A| = \frac{1}{|A^{-1}|}$ $\Rightarrow \mu_j = \frac{\sum_{i=1}^{n} p_{ij}^{(t)} x_i}{\sum_{i=1}^{n} p_{ij}^{(t)}}$

let $\Lambda_j = \Sigma_j^{-1}$ ← precision matrix

$p_{ij}^{(t)}\left(-\frac{d}{2}\ln(2\pi) + \frac{1}{2}\ln(|\Lambda_j|) - \frac{1}{2}(x_i-\mu_j)^T\Lambda_j(x_i-\mu_j)\right)$

take derivative wrt Λ_j

(if wrt Σ_j, will get the same result, but much harder)

$$\sum_{i=1}^{n} p_{ij}^{(t)}\left(\Sigma^T - \frac{1}{2}(x_i-\mu_j)(x_i-\mu_j)^T\right) = 0$$

$$\Rightarrow \Sigma_j = \frac{\sum_{i=1}^{n} p_{ij}^{(t)}(x_i-\mu_j)(x_i-\mu_j)^T}{\sum_{i=1}^{n} p_{ij}^{(t)}}$$

Variance collapse to 0 if $\mu_j = x_i$ for some i (i.e. one point is one cluster)